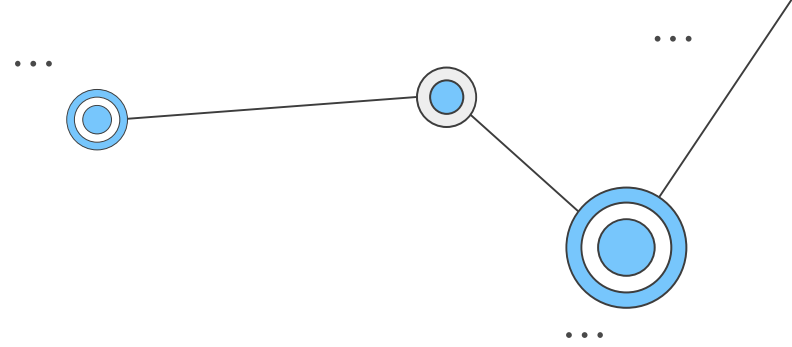
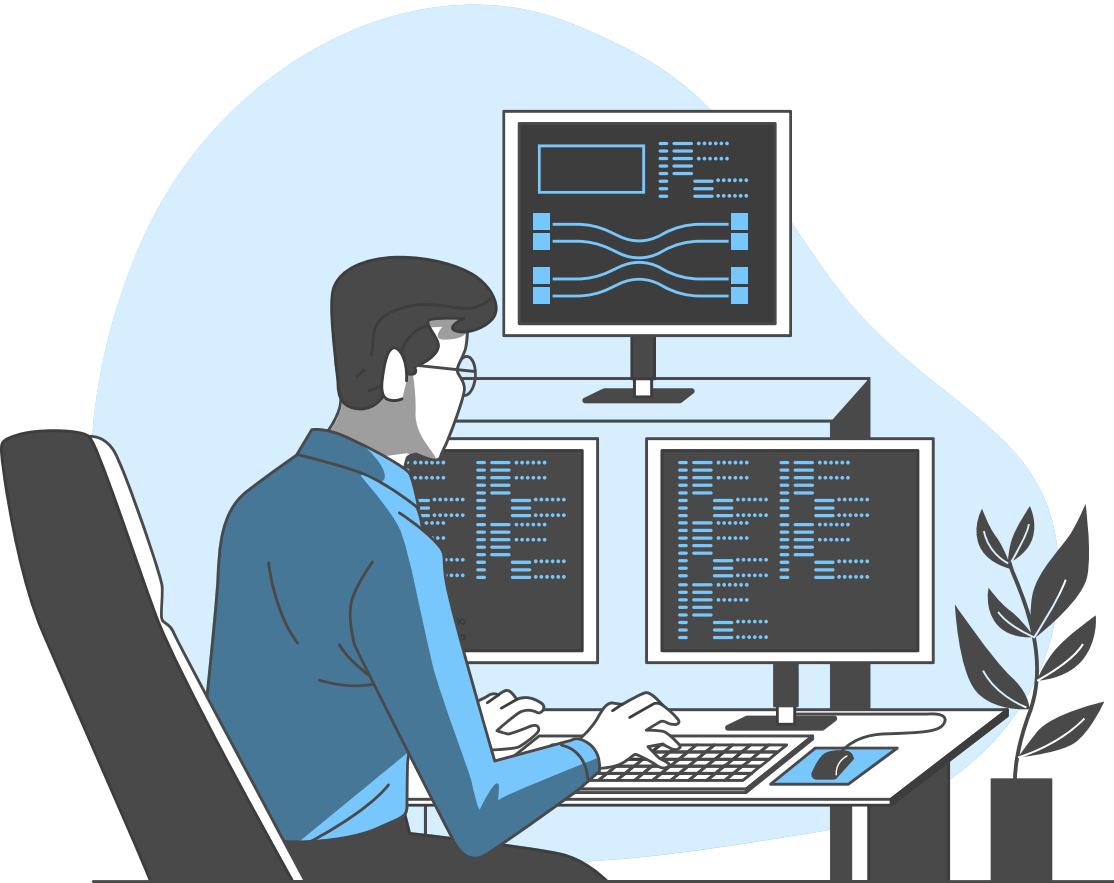
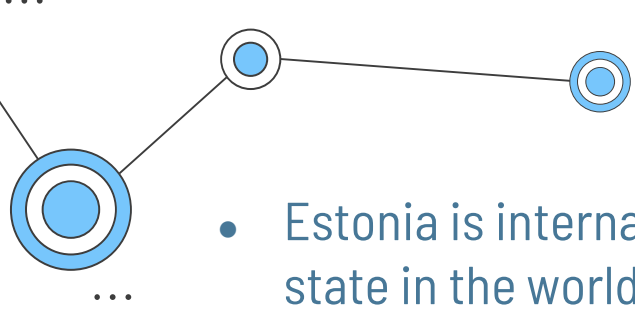




e- Identity

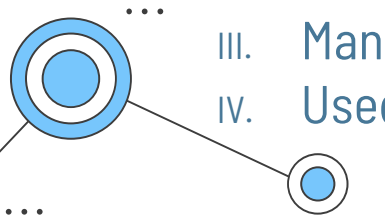
Presented by Aaryaka Timsina
Roll Number: 16
BPSG, 7th Semester



Estonia's e-Identity

- Estonia is internationally recognized as the most digitally advanced state in the world.
- At the core of its digital governance lies the e-Identity system, which allows citizens to securely access almost all public and private services online.

In overview,

- 
- I. e-Identity = state-issued digital identity for online authentication
 - II. Introduced in 2002 after independence from the Soviet Union
 - III. Mandatory for citizens; optional for residents & e-residents
 - IV. Used as a legal equivalent of physical identity



Core Components

1. Digital ID Card

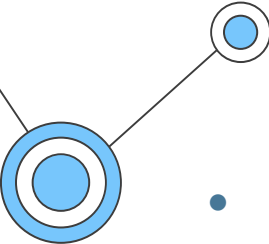
- Mandatory national identity card
- Contains cryptographic chips
- Used for authentication and digital signatures

2. Mobile-ID & Smart-ID

- Phone-based digital identity solutions
- More convenient than physical cards
- Widely used for banking and daily services

3. X-Road Data Exchange Platform

- Secure backbone connecting government databases
- Allows institutions to share data safely
- Eliminates duplication of information



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Uses and Application

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- Estonia's e-Identity is deeply integrated into citizens' everyday life.
- ...

Applications

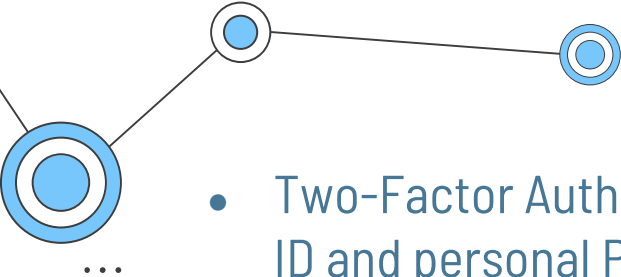
- e-Voting: Citizens can vote online in national elections
- e-Tax: Tax returns completed in minutes
- Digital Signatures: Legally binding and paperless
- e-Health: Access to medical records and prescriptions
- e-Business: Company registration in under 20 minutes



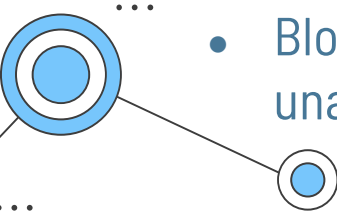
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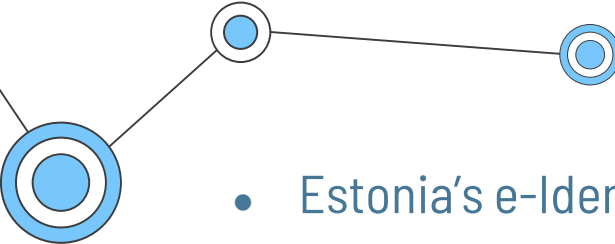
Over 99% of public services available online
Saves time, cost, and administrative burden

...



Security and Privacy Protection

- Two-Factor Authentication (PIN + ID): Requires both physical/digital ID and personal PIN to verify identity securely.
 - End-to-End Encryption: Ensures data is encrypted during transmission and can only be read by authorized parties.
 - Public Key Infrastructure (PKI): Uses cryptographic keys to authenticate users and create legally valid digital signatures.
 - Block chain (KSI): Protects data integrity by detecting any unauthorized alteration of government records.
- 



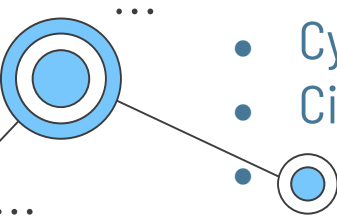
Global Significance and Lessons

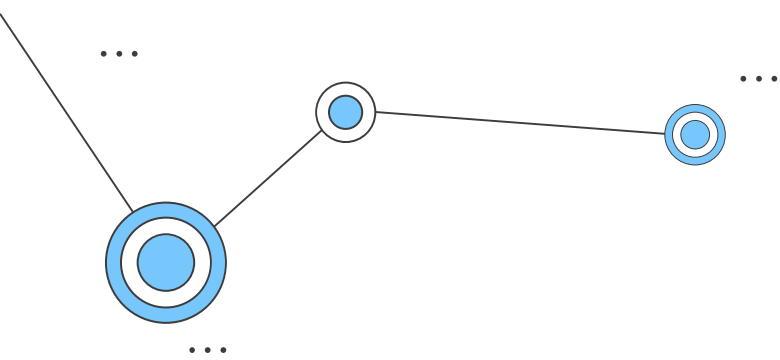
- Estonia's e-Identity system has become a global benchmark.

Global Impact

- Model for EU digital identity frameworks
- Adopted lessons by countries worldwide
- Basis for Estonia's e-Residency program

Key Lessons for Other Countries:

- 
- Strong legal framework is essential
 - Cybersecurity must be prioritized
 - Citizen trust and digital literacy matter
 - Technology must serve people, not control them



Thank you!

